

Strategic Interview Questions for Blue Robotics

Target Contact: Tony White (Applications Engineer) | **Goal:** Showcase Direct Value & Contribution Gaps

This document outlines targeted, technical questions designed to position you as a high-value contributor during your conversation with Blue Robotics. Each section pairs your distinct engineering milestones with a strategic, open-ended question to uncover their technical pain points.

1. Hardware Validation & Test Rig Design

Your Value Proposition: You have custom-engineered high-power thermal-fluid test rigs (handling heat fluxes up to 600 W/cm^2), operated under strict high-pressure/vacuum safety protocols, and authored formal hardware acceptance criteria.

*"At UCSD and Zap Energy, I custom-engineered high-power thermal-fluid test rigs, authored strict hardware acceptance criteria, and managed automated DAQ pipelines. As Blue Robotics expands into larger components or more complex thruster geometries, **where are your current testing bottlenecks?** Is there a need for someone to design more ruggedized, automated hardware test rigs or internal validation pipelines to accelerate product lifecycles?"*

2. Battery Intelligence & Autonomous Vehicle Software

Your Value Proposition: Developed a component-level Marine System Power Analysis Tool in Python, built solar-forecasting models, and formulated optimization algorithms (CVXPY) to keep batteries within safe operational envelopes.

*"During my Marine Energy Fellowship, I used Python to build component-level subsea power analysis tools and used CVXPY to keep lithium batteries in a safe State-of-Charge envelope over multi-week deployments. With the rollout of the BlueBoat and long-duration autonomous missions, **how is Blue Robotics tackling intelligent, predictive power budgeting?** I could see myself helping optimize battery duty-cycling or building telemetry data pipelines right into BlueOS."*

3. Thermal Management & Advanced Payload Integration

Your Value Proposition: Expert in thermal physics, modeling severe temperature gradients using COMSOL Multiphysics, evaluating effective thermal conductivity, and leading cross-functional working-fluid design reviews.

*"As customers integrate increasingly power-dense computing payloads (like edge AI processing or advanced sonar arrays) into your standard watertight enclosures, thermal dissipation becomes a massive challenge. Having modeled complex, severe-temperature multiphysics gradients in COMSOL, **is enclosure thermal management a constraint your engineering team is actively trying to solve***

for customers?"

4. Saltwater Materials Engineering & Biofouling

Your Value Proposition: Deep domain knowledge in saltwater falling-film evaporation, anti-fouling coatings, wettability boundaries, and microstructural analysis of scale morphology via SEM/EDX.

*"I've spent significant time analyzing saltwater scale dynamics, surface coatings, and hydrophobic materials to prevent degradation. As Blue Robotics gear is increasingly used for long-term, static infrastructure monitoring, **what are the biggest material or biofouling issues your customers report**, and do you see room to bring in more rigorous material characterization or surface-treatment validation?"*

5. The Closing Pitch

Your Value Proposition: Adaptable, multi-disciplinary engineer capable of bridging the gap between rigorous hands-on hardware prototyping and high-level software data analytics.

*"Tony, looking at the trajectory of Blue Robotics, you are bridging the gap between community-led open-source robotics and highly reliable industrial subsea tools. Given my background combining hands-on custom test fabrication with heavy data-driven modeling, **where do you see the biggest engineering gaps right now** where a multi-disciplinary engineer could hit the ground running?"*